

Technical report on
**Eco-Route Optimization System for Reducing Traffic Congestion
and Vehicle Emissions**

as the proposed solution for environmental issue of
Traffic Congestion and Vehicle Emissions

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ABSTRACT

Traffic congestion and vehicle emissions are major environmental and urban sustainability challenges associated with the increasing dependence on road transportation. Congested roads result in longer travel times, increased fuel consumption, vehicle idling and unnecessary emissions. Transport is also an important source of air pollution and greenhouse gas emissions, with transport-related pollution originating from both vehicle exhaust and non-exhaust sources such as brake and tyre wear.

This report proposes an **Eco-Route Optimization System**, an Engineering/IT-based solution designed to assist users in selecting environmentally preferable routes for their journeys. Instead of considering only the shortest distance or fastest travel time, the proposed system evaluates factors such as route distance, estimated travel time, traffic conditions, transportation mode, fuel consumption and estimated carbon dioxide emissions. Based on these parameters, the system can recommend a route that provides a suitable balance between travel efficiency and environmental impact.

The proposed system can be implemented as a responsive web-based application using HTML, CSS and JavaScript. Users can enter their source, destination and preferred mode of transportation, after which the system can compare available route options and provide information such as estimated travel time, distance, fuel consumption and emissions. The solution can also provide a sustainability or eco-score to help users make environmentally conscious transportation decisions.

The proposed approach supports the objectives of Sustainable Development Goal 11 (Sustainable Cities and Communities), particularly Target 11.2, which calls for safe, affordable, accessible and sustainable transport systems. The system also supports environmental management principles by enabling environmental performance to be considered, measured and improved systematically. ISO 14001:2026 provides a framework for organizations to manage environmental responsibilities and continually improve environmental performance.

The proposed system is intended to demonstrate how information technology can contribute to sustainable transportation by encouraging more efficient route selection, reducing unnecessary vehicle travel and supporting informed transportation choices.

INTRODUCTION

Transportation is an essential part of modern urban life, enabling people and goods to move between homes, workplaces, educational institutions and commercial areas. However, the rapid growth of urban populations and private vehicle usage has placed increasing pressure on road networks. When the number of vehicles exceeds the available road capacity, traffic congestion occurs, leading to slower movement and longer journey times.

Traffic congestion is not only a transportation problem but also an environmental concern. Vehicles operating in congested conditions may spend more time idling and frequently accelerating and braking, resulting in inefficient fuel use. Increased fuel consumption contributes to higher emissions, including carbon dioxide and other air pollutants. The World Health Organization identifies transport as an important source of air pollution and greenhouse gas emissions, with road transportation also contributing to pollution through sources such as brake and tyre wear.

Vehicle-related air pollution can affect both environmental quality and public health. Pollutants such as particulate matter and nitrogen oxides are associated with respiratory and cardiovascular health risks. Therefore, reducing unnecessary vehicle use, improving transportation efficiency and encouraging cleaner travel choices are important aspects of sustainable urban development.

The problem is particularly relevant to Sustainable Development Goal 11 (Sustainable Cities and Communities). Target 11.2 calls for safe, affordable, accessible and sustainable transport systems for all, while Target 11.6 focuses on reducing the environmental impact of cities.

Addressing traffic congestion traditionally involves measures such as road expansion, improved public transportation and traffic management. However, technological solutions can complement these approaches by helping individuals make more informed transportation choices. The availability of digital maps, route information and data-processing techniques creates opportunities to consider environmental factors alongside conventional transportation criteria.

This project therefore examines traffic congestion and vehicle emissions as a sustainability problem and investigates how an Engineering/IT-based approach can contribute to reducing their environmental impact.

PROBLEM STATEMENT

Rapid growth in urban vehicle usage has increased pressure on existing road infrastructure, resulting in frequent traffic congestion, particularly during peak hours. Congested roads cause vehicles to travel at inefficient speeds, remain idle for longer periods and undergo repeated acceleration and braking. These conditions increase travel time and fuel consumption and consequently contribute to increased vehicle emissions.

The problem is not limited to the overall number of vehicles on the road. The choice of route also influences the environmental impact of a journey. Two routes connecting the same locations may differ in distance, traffic density, travel time and fuel consumption. A route selected solely on the basis of minimum distance or travel time may therefore not always be the most environmentally preferable option.

At present, route selection by individual travellers is generally focused on reaching the destination quickly or conveniently. Environmental factors such as estimated fuel consumption and carbon emissions are less commonly considered as part of the user's route-selection decision. This creates an opportunity to use Information Technology to provide environmental information alongside conventional route-planning parameters.

Therefore, the problem addressed by this project is to assist travellers in selecting routes that reduce unnecessary fuel consumption and vehicle emissions while maintaining reasonable travel efficiency using an Engineering/IT-based system.

The project focuses on developing the concept of an Eco-Route Optimization System that considers both transportation efficiency and environmental impact when evaluating alternative routes. The objective is not to eliminate traffic congestion through infrastructure changes, but to provide a technological approach that can encourage more environmentally conscious route-selection decisions.

Causes of the Problem

The major factors contributing to traffic congestion and associated vehicle emissions include:

- Increasing private vehicle ownership and usage.
- Limited road capacity in densely populated urban areas.
- High traffic volumes during peak hours.
- Inefficient route selection by individual travellers.
- Vehicle idling and frequent acceleration and braking in congested conditions.
- Limited consideration of environmental impact during travel planning.

Effects of the Problem

The major consequences include:

- Increased travel time and delays.
- Higher fuel and energy consumption.
- Increased greenhouse gas emissions.
- Increased exposure to transportation-related air pollutants.
- Reduced fuel efficiency and higher transportation costs.
- Negative impacts on urban environmental quality and sustainability.

The problem therefore requires a solution that considers transportation efficiency together with environmental performance rather than treating route selection solely as a time- or distance-minimization problem.

SUSTAINABILITY ANALYSIS

The proposed Eco-Route Optimization System addresses traffic congestion and vehicle emissions from an environmental and sustainability perspective. Its sustainability can be evaluated using the Sustainable Development Goals (SDGs), ISO 14001, Life Cycle Assessment (LCA), and Environmental Impact Assessment (EIA).

Sustainable Development Goals

The project is primarily related to SDG 11 – Sustainable Cities and Communities, particularly Target 11.2, which promotes safe, affordable, accessible and sustainable transport systems.

It also supports:

- **SDG 13 – Climate Action:** Reduced fuel consumption can help reduce greenhouse gas emissions.
- **SDG 3 – Good Health and Well-being:** Lower vehicle emissions can contribute to better urban air quality and reduced exposure to harmful pollutants.

ISO 14001

ISO 14001:2026 provides a framework for organizations to manage environmental responsibilities and continually improve environmental performance.

The proposed system applies these principles by identifying fuel consumption and vehicle emissions as environmental aspects, setting the objective of reducing unnecessary travel and emissions, and evaluating environmental performance through estimated fuel and emission values.

Life Cycle Assessment (LCA)

ISO 14040:2006 provides the principles and framework for Life Cycle Assessment.

A simplified LCA perspective can be applied to the system:

<u>Stage</u>	<u>Environmental Impact</u>
Development	Energy and resources used for software development
Operation	Electricity used by devices, networks and servers
Use	Potential reduction in vehicle travel, fuel consumption and emissions
Maintenance	Energy required for updates and maintenance

The major potential benefit occurs during the use stage, if recommended routes reduce unnecessary vehicle travel and fuel consumption.

PROPOSED ENGINEERING/IT SOLUTION

Proposed Solution

The proposed solution is an Eco-Route Optimization System, a web-based application designed to help users select a comparatively sustainable route for their journey. The system considers both transportation efficiency and environmental impact instead of selecting a route based only on minimum distance or travel time.

The main objective is to identify a route that provides a suitable balance between travel time, traffic conditions, distance, fuel consumption and vehicle emissions.

Working Principle

The system follows a multi-step process:

User Input → Route Data → Route Analysis → Emission Estimation → Eco-Score → Route Recommendation

First, the user enters the source, destination and preferred transportation mode or vehicle type. The system then considers the available route alternatives and their characteristics, such as distance, estimated travel time and traffic level.

Each route is evaluated based on these parameters. Fuel consumption and estimated CO₂ emissions are then calculated using the distance and vehicle efficiency. Finally, the system assigns an eco-score to the available routes and recommends the route with the most suitable overall environmental and transportation performance.

Route Optimization

The key feature of the proposed system is that it does not treat the shortest route as automatically being the best route. A slightly longer route with significantly lower traffic may consume less fuel than a shorter route with heavy congestion.

The system can therefore use a weighted scoring approach. Each route receives scores for factors such as travel time, distance, traffic and estimated emissions. These individual scores are combined to produce an overall eco-score.

For example, the system could give greater importance to emissions and traffic when the objective is to prioritize environmental sustainability. The weights can be modified according to the requirements of the application.

A simplified representation is:

$$\text{Eco-Score} = w_1(\text{Travel Efficiency}) + w_2(\text{Traffic}) + w_3(\text{Emission Performance}) + w_4(\text{Distance})$$

where w_1 , w_2 , w_3 and w_4 represent the importance assigned to each factor.

The route with the most suitable overall score is then presented as the recommended eco-route.

Emission Estimation

The system estimates the environmental impact of a journey using distance and vehicle fuel efficiency.

A simplified calculation is:

$\text{Fuel Consumption} = \text{Distance} \div \text{Vehicle Fuel Efficiency}$

The estimated CO₂ emission can then be calculated using:

$\text{CO}_2 \text{ Emission} = \text{Fuel Consumed} \times \text{Fuel Emission Factor}$

The emission factor can be selected according to the fuel type using a recognized environmental reference. Since actual emissions vary depending on vehicle condition, driving behaviour, traffic and other factors, the values produced by the prototype are treated as estimates rather than certified measurements.

System Architecture

The system consists of three main layers.

The presentation layer provides the user interface through which users enter trip details and view route recommendations. It can be developed using HTML and CSS.

The processing layer uses JavaScript to compare routes, perform emission calculations and generate eco-scores. This layer contains the main decision-making logic of the prototype.

The data layer contains route and vehicle information required for calculations. For the academic prototype, predefined route data can be used. A future version could obtain real-time traffic and route information through mapping and traffic APIs.

Thus, the overall architecture can be represented as:

User → Web Interface → Route & Vehicle Data → Processing and Calculation → Eco-Score → Recommendation

Example of Operation

Suppose two routes are available between the same locations. One route is shorter but has heavy traffic, while the other is slightly longer but has low traffic.

A conventional route planner may select the shorter route. The proposed system, however, evaluates both routes using traffic conditions, estimated fuel consumption and emissions. If the longer route results in significantly lower fuel consumption and emissions, it can receive a better eco-score and be recommended as the sustainable alternative.

This allows the user to understand not only which route is available, but also why one route may be environmentally preferable.

Technologies Used

The prototype can be developed using basic web technologies. HTML is used to structure the webpages, CSS is used for layout and responsive design, and JavaScript performs route comparison, calculations, eco-score generation and interactive functions.

For a future full-scale implementation, the system could incorporate GPS, digital mapping, real-time traffic information and live vehicle data. These additions would allow the recommendations to become more dynamic and accurate.

Expected Functionality

The proposed system is designed to allow users to enter trip details, compare alternative routes, view estimated travel time and distance, calculate fuel consumption and CO₂ emissions, obtain an eco-score and receive a recommended route.

The solution therefore demonstrates how Information Technology can be used to integrate transportation planning and environmental decision-making within a single system.

FEASIBILITY ANALYSIS

The feasibility of the proposed Eco-Route Optimization System is evaluated based on its technical, economic, operational and environmental aspects.

Technical Feasibility

The system can be developed using HTML, CSS and JavaScript. Route comparison, emission calculation and eco-score generation can be performed using JavaScript. Predefined route and vehicle data can be used for the prototype, so a database or backend is not required.

Economic Feasibility

The prototype requires minimal cost since it uses freely available web-development tools and does not require specialized hardware. A future large-scale version may require additional costs for hosting, mapping services and real-time traffic data.

Operational Feasibility

The website is designed to be simple and user-friendly. Users enter their trip details and receive route comparisons and environmental information. It can be used by students, daily commuters and other travellers without specialized technical knowledge.

Environmental Feasibility

The system can encourage users to choose routes with lower estimated fuel consumption and emissions. However, the actual environmental benefit depends on factors such as user adoption, traffic conditions and the accuracy of the available data.

Overall Feasibility

Overall, the proposed Eco-Route Optimization System is feasible as an academic prototype because it requires limited resources, uses readily available technologies and can be implemented as a responsive website. Its functionality can also be expanded in the future through real-time traffic data, mapping services and more advanced emission models.

EXPECTED ENVIRONMENTAL BENEFITS

The proposed Eco-Route Optimization System is designed to balance environmental impact and travel efficiency while helping users make better route choices.

Reduced Travel Time and Delays

By considering traffic conditions along with distance, the system can help users avoid heavily congested routes. This can reduce unnecessary delays and improve the overall efficiency of a journey.

Reduced Fuel Consumption

Avoiding highly congested or unnecessarily long routes can reduce idling, repeated acceleration and excessive travel distance, potentially lowering fuel consumption.

Reduction in Vehicle Emissions

Reduced fuel consumption can lead to lower CO₂ emissions and other vehicle-related environmental impacts. The system provides estimated emission values so users can compare the environmental performance of different routes.

Better Balance Between Time and Sustainability

The system does not focus only on the fastest or shortest route. Instead, it considers travel time, distance, traffic and estimated emissions together. This allows the system to recommend a route that provides a practical balance between reaching the destination efficiently and reducing environmental impact.

Increased Environmental Awareness

Displaying fuel consumption, emissions and eco-scores alongside travel time can help users understand the environmental consequences of their route choices and encourage more sustainable travel behaviour.

Overall, the system aims to provide efficient and environmentally conscious route recommendations, while ensuring that sustainability does not come at the cost of unreasonable travel delays.

JUSTIFICATION OF PROPOSED SOLUTION

The Eco-Route Optimization System is proposed because conventional route selection generally focuses on reaching the destination in the shortest possible time or distance, while environmental factors are given less importance. Considering traffic conditions, fuel consumption and estimated emissions along with travel time can provide users with a more sustainable basis for choosing a route.

The solution is also practical because it can be implemented using basic web technologies such as HTML, CSS and JavaScript. The prototype does not require specialized hardware or complex infrastructure and can operate using predefined route and vehicle data.

Another advantage is that the system considers the trade-off between travel efficiency and environmental impact. It does not simply recommend the route with the lowest emissions if that route requires an unreasonable increase in travel time. Instead, multiple factors are evaluated together to identify a comparatively efficient and environmentally preferable route.

The system can also improve environmental awareness by showing users estimated fuel consumption, CO₂ emissions and an eco-score for different routes. This allows users to understand how their transportation choices can affect environmental performance.

Therefore, the proposed system is justified as a simple, feasible and scalable Engineering/IT solution that can complement existing transportation measures by encouraging more informed and environmentally conscious route-selection decisions.

CONCLUSION

Traffic congestion and vehicle emissions are significant challenges to sustainable urban transportation. Congested roads increase travel time, fuel consumption and vehicle emissions, making efficient transportation planning important for reducing environmental impact.

This project proposed an Eco-Route Optimization System that considers travel time, distance, traffic conditions, fuel consumption and estimated CO₂ emissions when comparing alternative routes. By combining these factors into an eco-score, the system can recommend a route that provides a reasonable balance between travel efficiency and environmental sustainability.

The proposed web-based system is technically and economically feasible as an academic prototype using HTML, CSS and JavaScript. It can also be extended in the future by incorporating real-time traffic data, GPS and mapping services for large scale implementation.

Overall, the project demonstrates how Engineering and Information Technology can support sustainable transportation by helping users make more informed route-selection decisions and encouraging environmentally conscious travel behaviour.

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